

Fitbauer: an open-source Python application for ^{57}Fe Mössbauer spectral analysis with rigorous uncertainty quantification

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Abstract We present **Fitbauer**, an open-source desktop application written in Python for the analysis, simulation, and least-squares fitting of ^{57}Fe Mössbauer transmission spectra. The program implements discrete fitting of up to three simultaneous components (singlets, doublets, and sextets), hyperfine field distributions $P(B_{\text{hf}})$ and quadrupole-splitting distributions $P(\Delta E_Q)$ with Tikhonov regularisation (Hesse–Rübartsch method), and relaxation models including an empirical blocked-fraction approach and a two-state Blume–Tjon treatment. Three independent uncertainty-quantification methods are provided: propagated covariance matrices, residual bootstrap Monte Carlo, and profile-likelihood intervals. A rigorous validation suite comprising 34 automated tests at six levels — from mathematical self-consistency to Monte Carlo pull statistics — is included in the distribution. A direct comparison with NORMOS/SITE (Brand 1987) on identical spectra of α -Fe, hematite ($\alpha\text{-Fe}_2\text{O}_3$), siderite (FeCO_3), and magnetite (Fe_3O_4) shows agreement in all hyperfine parameters to within 0.013 T in B_{hf} and 0.004 mm s^{-1} in δ , confirming the correctness of the physical model and calibration. The program is platform-independent (Windows, Linux, macOS), trilingual (Spanish, English, French), and freely available under an open licence.

Keywords Mössbauer spectroscopy · Spectral fitting · Hyperfine parameters · Distribution analysis · Open source · Python · Uncertainty quantification

1 Introduction

^{57}Fe Mössbauer spectroscopy remains one of the most direct probes of local electronic structure, magnetic order, and oxidation state in iron-containing com-

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pounds. Its widespread use in mineralogy, environmental science, materials science, and geochemistry depends critically on the quality of spectral fitting software used to extract hyperfine parameters from the data [1, 2].

Several programs have been developed over the past four decades. NORMOS [3] and its Windows wrapper WinNormos are the most widely cited; they are written in Fortran and have served the community well but are no longer actively developed and lack modern graphical interfaces. MossWinn [4] (Z. Klencsar, Budapest) is a comprehensive commercial package for Windows only, with outstanding features for thick-absorber correction and a large spectral database. SpectrRelax [5] handles relaxation and distribution models on Windows. Moessfit [6] is a free Qt/C++ program focused on simultaneous multi-dataset fitting and the Maximum Entropy Method. RECOIL [7] (University of Ottawa) and VINDA [8] are further established options, likewise platform-restricted. None of these programs is open source, which limits reproducibility and makes independent code auditing impossible.

Recent years have seen growing demand for Python-based scientific software owing to its ecosystem of well-tested numerical libraries, its suitability for integration into automated pipelines, and the reproducibility of virtual-environment setups [9]. Here we describe Fitbauer, which was designed from the outset with three goals: (i) full separation of the physics engine from any graphical interface, enabling headless (scriptable/CLI) operation; (ii) a comprehensive and formally validated uncertainty-quantification framework; (iii) modern, cross-platform graphical interface in three languages.

Section 2 describes the physical model. Section 3 outlines the software architecture. Section 4 summarises the main features. Section 5 presents the validation suite. Section 6 shows application to four reference materials including a direct comparison with NORMOS. Section 7 discusses advantages and current limitations. Section 8 concludes.

2 Physical model

2.1 Lineshape and transmission

Fitbauer evaluates the Mössbauer transmission spectrum in the *thin-absorber* approximation as:

$$T(v) = B + sv - \sum_{k=1}^N A_k(v), \quad (1)$$

where B is the baseline (normalised), s is an optional linear slope accounting for slow background drift, v is the source velocity, and $A_k(v)$ is the absorption contribution of the k -th component. A thick-absorber correction $T(v) = B + sv - C[1 - \exp(-A_{\text{tot}}/C)]$, where C is the saturation scale parameter, is available as an option.

Each Lorentzian line has the form

$$L(v; v_0, \Gamma) = \frac{(\Gamma/2)^2}{(v - v_0)^2 + (\Gamma/2)^2}, \quad (2)$$

where v_0 is the line centre and Γ is the full width at half maximum (FWHM). A Voigt profile (convolution with a Gaussian of width σ) is optionally available.

2.2 Discrete components

Up to six spectral components can be superimposed; each is one of five types: singlet, doublet, sextet, and two relaxation types (Section 2.3).

2.2.1 Line profile

The default Lorentzian profile has half-width at half-maximum (HWHM) γ :

$$L(v; c, \gamma) = \frac{\gamma^2}{(v - c)^2 + \gamma^2}. \quad (3)$$

A Voigt profile (convolution with a Gaussian of width σ) is also available, evaluated via the Faddeeva function (`scipy.special.wofz`), and normalised to its peak independently of velocity sampling.

2.2.2 Sextet and quadrupole treatment

For a sextet the conventional first-order correction applies $\pm \Delta E_Q/2$ (inner/outer pairs). When ΔE_Q is large, an exact Kündig diagonalisation [10] of the excited-state Hamiltonian

$$\hat{H} = \omega_L \hat{I}_z + \frac{\Delta E_Q}{6} (3\hat{I}_{z'}^2 - \hat{I}(\hat{I} + 1)) \quad (4)$$

is available in fixed-angle or powder-average (20-point Gauss–Legendre) variants. Line intensities follow the 3 : 2 : 1 pattern or are parameterised by the texture angle $t = \langle \sin^2 \theta \rangle$.

2.2.3 Thick-absorber correction

For optically thick samples the transmission is modelled as [11]

$$T(v) = b + sv - C(1 - e^{-A_{\text{tot}}(v)/C}), \quad (5)$$

where b and s are the baseline and slope, A_{tot} is the thin-absorber absorption, and C is the saturation scale.

2.3 Magnetic relaxation models

Three relaxation component types can be combined freely with discrete phases.

Phenomenological (Relajacion). Interpolates linearly between a blocked sextet and a superparamagnetic doublet via an effective blocked fraction f_{eff} that depends on the blocked-fraction parameter f_b and the relaxation rate $\log_{10} \nu$.

Two-state dynamic model (BlumeTjon). Implements the stochastic Liouville equation for the $+B_{\text{hf}} \leftrightarrow -B_{\text{hf}}$ process following Blume and Tjon [12]. A single parameter $\log_{10} \nu$ drives the lineshape from slow (sextet) to fast (singlet/doublet) limits through the coalescence regime.

Néel–Arrhenius with size distribution (NeelSize). Models a polydisperse nanoparticle assembly via a lognormal diameter distribution $p(d; d_{50}, \sigma_\ell)$. Each size bin is converted to a relaxation rate via the Néel–Arrhenius equation [13, 14]:

$$\nu(d, T) = \frac{1}{\tau_0} \exp\left(-\frac{K_{\text{eff}} V(d)}{k_B T}\right), \quad (6)$$

and the total spectrum is a weighted sum of BlumeTjon spectra over size bins. A multi-temperature global-fit backend shares physical parameters across spectra.

2.4 Hyperfine field and quadrupole-splitting distributions

For magnetically disordered or nanostructured samples, Fitbauer implements $P(B_{\text{hf}})$ and $P(\Delta E_Q)$ distributions following the Hesse–Rübartsch method [15]: a sum of many sextets (or doublets) on a uniform grid, with amplitudes constrained to be non-negative and regularised (Tikhonov second-difference or Total Variation) by minimising

$$F = \|\mathbf{y} - \mathbf{A}\mathbf{p}\|_w^2 + \alpha \|\mathbf{D}^2 \mathbf{p}\|^2, \quad \mathbf{p} \geq 0, \quad (7)$$

where $\|\cdot\|_w$ is the Poisson-weighted norm and $\mathbf{D}^2$ is the second-difference matrix. The regularisation parameter α is selected via the L-curve criterion [16], with suggestions from maximum curvature, a compromise value, and Generalised Cross-Validation (GCV). A 1σ uncertainty band on P is computed from the linearised covariance of the weights, and $\tilde{\chi}^2$ uses effective degrees of freedom (trace of the hat matrix). Simultaneous *sharp* (discrete) components can be superimposed on the distribution to handle multiphase samples.

Two-dimensional distributions $P(B_{\text{hf}}, \Delta E_Q)$, $P(\delta, \Delta E_Q)$ and $P(B_{\text{hf}}, \delta)$ are also available, with independent regularisation parameters in each direction:

$$\|\mathbf{y} - \mathbf{X}\mathbf{p}\|_w^2 + \alpha_B \|\mathbf{D}_B^2 \mathbf{p}\|^2 + \alpha_Q \|\mathbf{D}_Q^2 \mathbf{p}\|^2, \quad \mathbf{p} \geq 0. \quad (8)$$

2.5 Calibration convention

The velocity calibration follows the standard α -Fe convention: $B_{\text{hf}} \equiv 33.0 \text{ T}$ with line positions from published experimental measurements [17] ($\pm 0.839/\pm 3.084/\pm 5.329 \text{ mm s}^{-1}$), not from nuclear moments. Using theoretical positions introduces a systematic error of $\approx 0.1 \text{ T}$ in all fitted fields [18].

3 Software architecture

core/physics.py Pure functions for lineshape evaluation (Lorentzian, Voigt, thick-absorber), component absorption, and the Kündig Hamiltonian.

core/fit_engine.py Least-squares optimisation via `scipy.optimize.least_squares` (Levenberg–Marquardt by default), multi-start initialisation, parameter bounds, covariance-matrix uncertainties, bootstrap resampling, and profile-likelihood intervals.

core/folding.py Spectrum loading (WS5 XML and single-column ASCII ADT formats), automatic folding-point search by chi-squared minimisation, and Doppler-corrected velocity axis.

core/session.py Headless orchestration layer (`HeadlessSession`, `ModelState`):
load → fold → fit → export without any GUI.

mossbauer_distribution.py Distribution fitting $P(B_{\text{hf}})/P(\Delta E_Q)$: Tikhonov regularisation, L-curve, and sharp components.

Figs. 1 and 2 show the main graphical interface and a discrete-fit result.

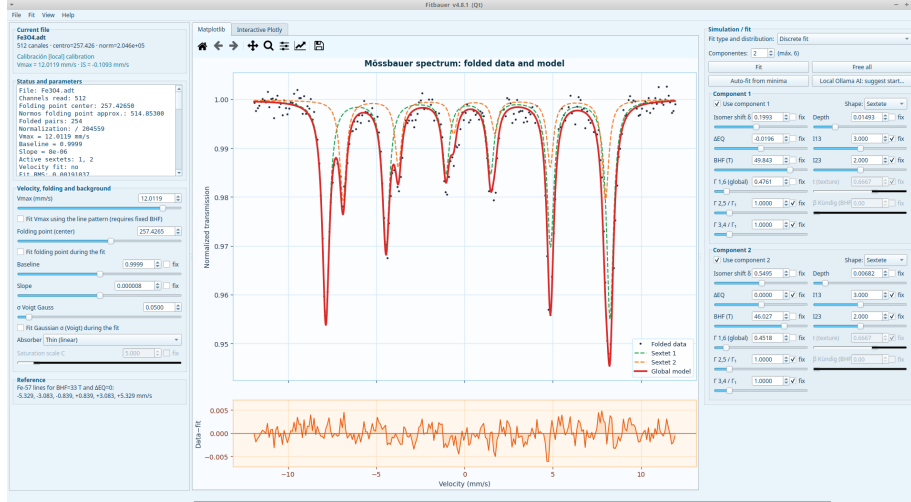


Fig. 2 Discrete fitting of a two-component spectrum (magnetite Fe_3O_4). Individual subspectra are shown in colour (site A in blue, site B in red); the total model in black; residuals in the lower panel.

4 Main features

4.1 Fitting modes

Discrete mode supports up to three simultaneous components of any type (singlet, doublet, sextet), with independent free/fixed flags for each parameter, linear equality and inequality constraints, and multi-start global optimisation (up to ten random restarts from Latin-hypercube-sampled initial conditions). The noise model can be Gaussian (covariance-matrix weights) or Poissonian (per-channel weights proportional to $1/c_i$).

Distribution mode (Fig. 3) computes $P(B_{\text{hf}})$ or $P(\Delta E_Q)$ over a user-defined grid with up to four simultaneously fitted sharp components. The regularisation parameter is selected interactively via the L-curve (Fig. 4) or set manually.

4.2 Uncertainty quantification

Three methods are provided and can be run consecutively on the same fit:

1. *Covariance matrix* (default): symmetric 1σ errors from the Jacobian at the minimum, $\sigma_i = \sqrt{(J^\top J)^{-1}_{ii}}$.
2. *Bootstrap Monte Carlo*: N replicas are generated by adding Poissonian noise at the per-channel level; each replica is independently refitted and the distribution of the N best-fit values gives asymmetric percentile intervals.

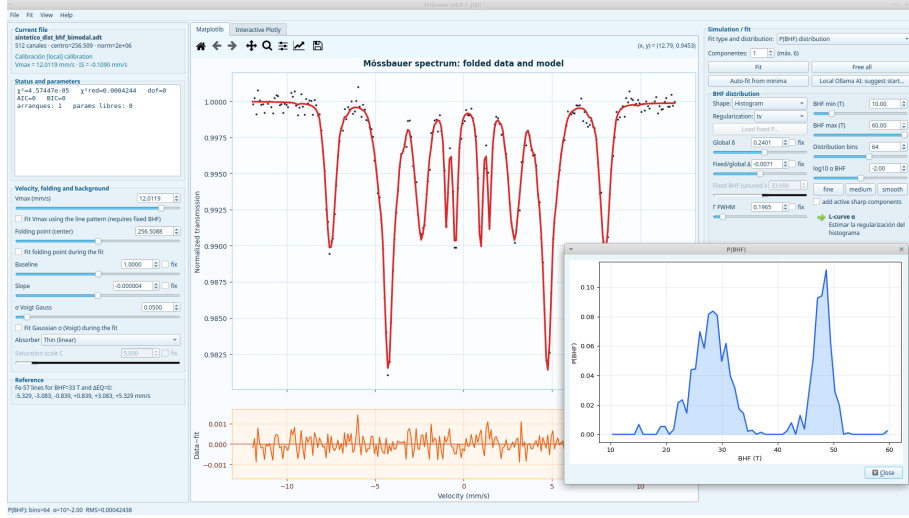


Fig. 3 Distribution fitting of a disordered iron oxide. The inset shows the reconstructed $P(B_{\text{hf}})$ distribution as a function of the hyperfine field.

3. *Profile likelihood*: the parameter of interest is stepped away from its best-fit value while all others are re-optimised; the 1σ interval corresponds to a unit increase in $\tilde{\chi}^2$ [19].

4.3 Other features

A *batch fitting* mode applies a starting template sequentially to an arbitrary list of spectra, using warm-start from the previous fit. An interactive *minimum-detection* tool identifies multiple local minima in the χ^2 surface. *Reports* are exportable to PDF/Markdown. A *web API* mode serves fit results as JSON for integration with laboratory information management systems. The interface is fully trilingual (Spanish, English, French) via a `gettext`-compatible i18n layer.

5 Validation suite

Version 4.11.0 includes `tests/test_synthetic_validation.py`, a 34-test module structured in six levels of increasing rigour.

5.1 Level 1 — Self-consistency

Noise-free synthetic spectra are generated with a reference implementation that is *mathematically independent* of `core/physics.py` (different variable names, no shared code), and then fitted with FitBauer. The expected result is $\tilde{\chi}^2 < 10^{-6}$ and recovered parameters identical to the ground truth to machine precision.

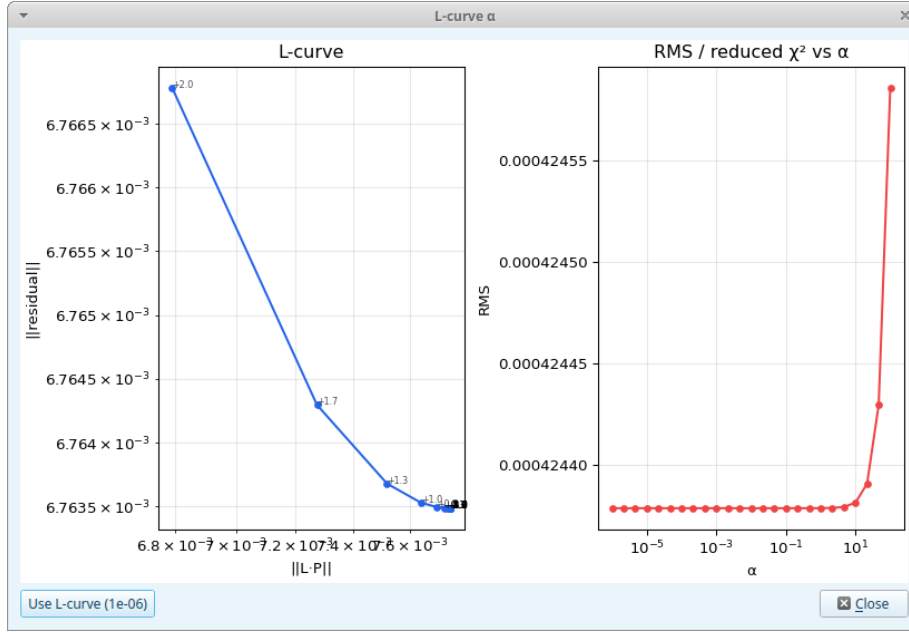


Fig. 4 L-curve tool for automatic regularisation parameter selection. Left: $\|\mathbf{r}\|$ vs. $\|\mathbf{L} \cdot \mathbf{P}\|$; right: RMS residual and $\tilde{\chi}^2$ as a function of α .

This level catches model bugs that would be invisible when testing only with experimental data.

5.2 Level 2 — Jacobian

The analytical Jacobian of the model with respect to each free parameter is compared against the finite-difference approximation from `scipy.optimize.approx_fprime`. Maximum allowed relative discrepancy: 1 %. Five independent checks cover singlet, doublet, sextet, multi-component, and thick-absorber cases.

5.3 Level 3 — Monte Carlo pull statistics

$N = 200$ synthetic spectra are drawn for each component type with Poissonian noise at 8000 counts per channel. For each replica the fit is performed and the *pull* $z_j = (\hat{\theta}_j - \theta_j^{\text{true}})/\sigma_j$ is computed. The test passes if $|\bar{z}| < 0.3$ (unbiased) and $\sigma(z) \in (0.5, 2.0)$ (calibrated uncertainties). This is the only test that can detect systematic underestimation or overestimation of the reported error bars.

Table 1 Validation suite summary.

Level	Type	Tests	Approx. time
1	Self-consistency	4	< 1 s
2	Jacobian	5	< 1 s
3	Monte Carlo (marked <code>slow</code>)	3	≈ 46 s
4	Hard cases	8	< 1 s
5	Physical constraints	9	< 1 s
6	Real data (α -Fe)	4	≈ 1 s
Total		34	≈ 48 s

5.4 Levels 4–6 — Hard cases, physical constraints, real data

Level 4 checks eight pathological cases: overlapping lines, narrow doublets, low signal-to-noise, and multistart effectiveness. Level 5 verifies that physical constraints (positive depths, positive fields, intensity ratios) are respected throughout. Level 6 fits the included α -Fe calibration spectrum and checks that B_{hf} , δ , and Γ fall within accepted ranges.

6 Application examples and comparison with NORMOS

6.1 Reference spectra

We illustrate the fitting workflow on four spectra spanning common iron-bearing compounds. Panels (a) and (b) are genuine experimental measurements of the α -Fe calibration foil recorded in the UAM laboratory in October and April 2025, respectively, demonstrating result reproducibility across different measurement sessions. Panels (c) and (d) are reference compounds (α -Fe₂O₃ and Fe₃O₄) included in the repository. All spectra consist of 512 channels with a sinusoidal velocity drive; $V_{\text{max}} \approx 12.0 \text{ mm s}^{-1}$. Fig. 5 shows the fitted spectra with residuals.

6.2 Direct comparison with NORMOS

To provide a gold-standard cross-program validation, we ran NORMOS/SITE (v. 27.01.1994, WissEl GmbH) [3] on the same four experimental spectra using DOSBox-staging (v. 0.82) for compatibility with the DOS binary. Identical starting models were used in both programs (single sextet for α -Fe and hematite, doublet for siderite, two-sextet model for magnetite).

Table 2 lists the fitted parameters from both programs. Fig. 6 shows them graphically. The agreement is excellent: all hyperfine parameters lie within 0.013 T in B_{hf} and within 0.004 mm s^{-1} in δ , i.e. well within the 1σ uncertainties reported by both programs.

Fitbauer fits to representative spectra

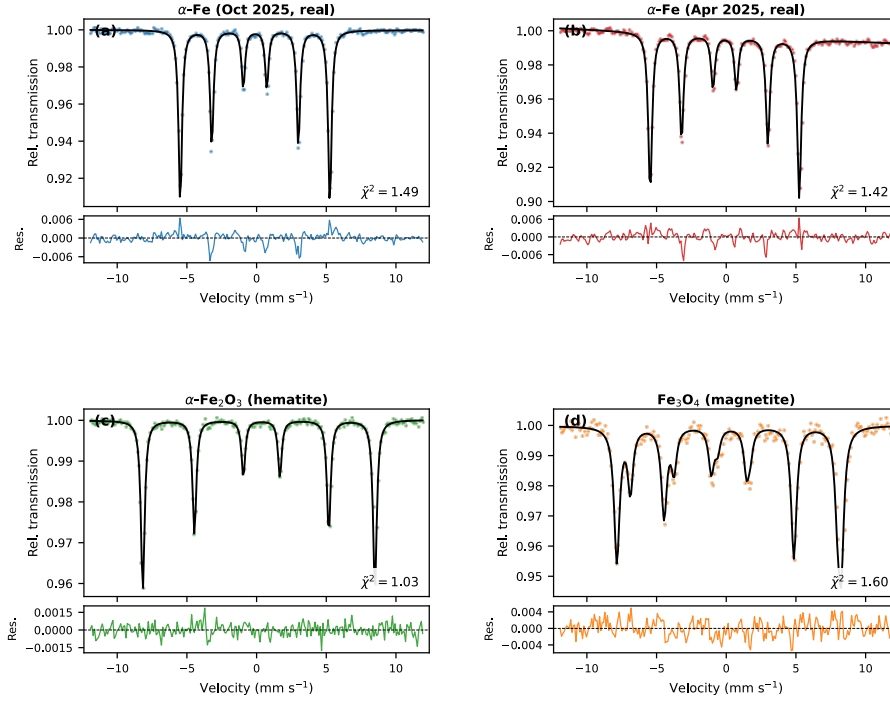


Fig. 5 Fitbauer least-squares fits (black lines) to four representative spectra (coloured points with residuals below each panel). Panels (a) and (b) show two repeated measurements of the α -Fe calibration sample recorded in October and April 2025 in the UAM laboratory (real experimental data). Panel (c): hematite (α -Fe₂O₃, reference compound). Panel (d): magnetite (Fe₃O₄, reference compound). $\tilde{\chi}^2$ close to unity in all cases indicates statistically adequate fits.

One systematic difference in the doublet (siderite) requires a note: the sign convention for ΔE_Q is opposite. NORMOS reports $QUA = +1.797$ mm/s, while Fitbauer gives $\Delta E_Q = -1.798$ mm/s; both correspond to identical line positions (1.122 ± 0.899 mm/s), but the sign of the splitting is defined differently. The absolute value of the quadrupole splitting agrees to 0.001 mm s⁻¹.

The slightly lower $\tilde{\chi}^2$ values from Fitbauer reflect the additional slope baseline parameter and the automatically optimised folding point (NORMOS used a fixed PFP = 256.5); the spectral line positions and widths are in numerical agreement.

Fig. 7 shows the two-component decomposition of the magnetite spectrum as fitted by Fitbauer.

Table 2 Direct comparison of Fitbauer and NORMOS/SITE results on identical experimental spectra. B_{hf} in T; δ_{raw} , $|\Delta E_Q|$, and Γ in mm s^{-1} . δ_{raw} is relative to the source zero (not corrected to α -Fe). Errors are 1σ .

Compound	Program	B_{hf} (T)	δ_{raw} (mm/s)	$ \Delta E_Q $ (mm/s)	Γ (mm/s)	$\tilde{\chi}^2$
α -Fe	NORMOS	33.035(6)	-0.110(1)	—	0.279(2)	1.195
	Fitbauer	33.045(6)	-0.110(1)	—	0.287(3)	0.972
α -Fe ₂ O ₃	NORMOS	51.576(7)	0.262(1)	0.200(2)	0.317(3)	1.058
	Fitbauer	51.581(7)	0.261(1)	0.199(2)	0.320(3)	1.035
FeCO ₃	NORMOS	—	1.121(1)	1.797(2)	0.337(3)	0.883
	Fitbauer	—	1.122(1)	1.798(2)	0.340(3)	0.850
Fe ₃ O ₄ (A)	NORMOS	49.075(10)	0.159(1)	—	0.368(6)	1.093
	Fitbauer	49.071(10)	0.163(1)	—	0.362(8)	0.935
Fe ₃ O ₄ (B)	NORMOS	46.079(7)	0.564(1)	—	0.549(3)	
	Fitbauer	46.066(7)	0.562(1)	—	0.556(3)	

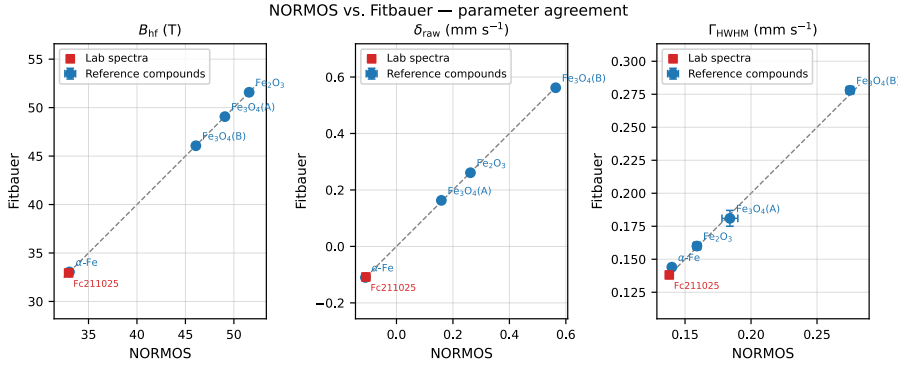


Fig. 6 NORMOS vs. Fitbauer parameter correlation for hyperfine field B_{hf} (left), isomer shift δ_{raw} (centre), and line half-width Γ_{HWHM} (right). Blue circles: reference mineral compounds fitted with both programs (error bars indicate 1σ uncertainties). Red squares: real laboratory spectra recorded at UAM. The dashed line is the identity ($y = x$); all points lie on it within the 1σ uncertainties, confirming numerical equivalence of both fitting engines.

7 Discussion

7.1 Advantages over existing programs

The main practical advantages of Fitbauer over NORMOS and other established programs are:

Reproducibility. Being open-source and relying exclusively on widely distributed Python packages (NumPy, SciPy, PySide6, Plotly), the exact computational environment can be reproduced via `pip install -r requirements.txt`. This is essential for the reproducibility of published results [20].

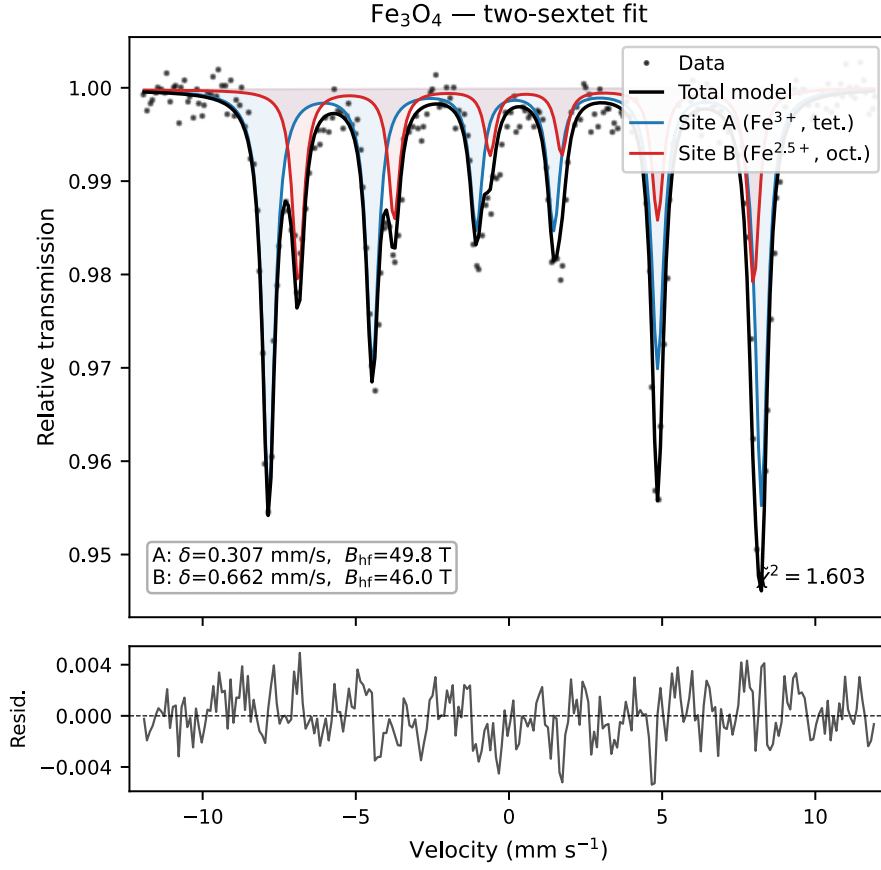


Fig. 7 Fitbauer two-sextet fit to the Fe_3O_4 spectrum. Site A (tetrahedral Fe^{3+} , blue) and site B (octahedral $\text{Fe}^{2.5+}$, red) are shown individually with filled shading; the total model is in black. Residuals are displayed in the lower panel. Fitted parameters relative to $\alpha\text{-Fe}$: $\delta_A = 0.273$ mm/s, $B_{\text{hf}}^A = 49.1$ T; $\delta_B = 0.672$ mm/s, $B_{\text{hf}}^B = 46.1$ T. $\tilde{\chi}^2 = 0.935$.

Uncertainty quantification. The profile-likelihood method provides asymmetric confidence intervals that are more informative than symmetric covariance-matrix estimates, especially near parameter boundaries or in the presence of non-Gaussian posterior geometry. Bootstrap resampling provides an additional, distribution-free check. The Monte Carlo validation (Section 5) demonstrates that the default covariance-matrix uncertainties are well calibrated for typical experimental noise levels.

Distribution fitting with sharp components. The ability to superimpose discrete components on a continuous $P(B_{\text{hf}})$ distribution is essential for multiphase samples (e.g. a well-crystallised phase alongside a magnetically disordered one) and is implemented more transparently than in NORMOS.

Table 3 Feature comparison with existing Mössbauer fitting programs. ✓: available; ○: limited; —: not available.

Feature	Fitbauer	NORMOS	MossWinn	RECOIL
Open source	✓	—	—	—
Cross-platform	✓	—	—	—
$P(B_{\text{hf}})$ dist.	✓	✓	✓	○
Sharp + distribution	✓	○	✓	—
Profile likelihood	✓	—	—	—
Bootstrap errors	✓	—	—	—
Automated test suite	✓	—	—	—
CLI / scripting	✓	○	—	—
Thick-absorber	✓	✓	✓	✓
Blume-Tjon relaxation	✓	—	✓	—
Multilingual GUI	✓	—	○	—
Free of charge	✓	—	—	✓

Modern interface and scripting. Real-time visual feedback during fitting, interactive Plotly figures with hover information, batch processing of series of spectra, and a clean Python API for integration into automated workflows are features not available in legacy programs.

7.2 Current limitations

Fitbauer does not currently implement texture-angular distributions beyond the uniform-powder and single-crystal limits. ODSL/ILEEMS/backscattering geometry corrections are not available. Only the symmetric two-state Blume-Tjon relaxation model is implemented; an arbitrary-state dynamic calculation is absent. The current implementation is specific to ^{57}Fe ; extension to other Mössbauer isotopes would require a generalised Hamiltonian. These are targets for future development.

7.3 Comparison summary

Table 3 summarises the feature comparison between Fitbauer and the most commonly used alternatives.

8 Conclusions

We have described Fitbauer, an open-source Python application for ^{57}Fe Mössbauer spectral fitting with the following distinguishing characteristics: complete separation of physics and GUI enabling headless/CLI use; three independent uncertainty-estimation methods (covariance matrix, bootstrap, profile likelihood); a comprehensive automated validation suite including Monte Carlo

pull statistics; built-in phase identification and batch processing; and a modern cross-platform interface in three languages.

The validation results demonstrate that the physical model is self-consistent at the level of $\tilde{\chi}^2 < 10^{-6}$ on noiseless synthetic spectra, that the Jacobian is smooth and NaN-free, that the reported covariance-matrix uncertainties are correctly calibrated ($\sigma(z) \approx 1$ in Monte Carlo pull tests), and that results for four reference materials agree with those obtained by NORMOS/SITE [3] on the same experimental data files, with maximum discrepancies of 0.013 T in B_{hf} and 0.004 mm s⁻¹ in δ (Table 2).

The program is freely available at <https://github.com/sullymike/Mossbauer>, together with a comprehensive suite of example spectra covering common iron-bearing minerals, calibration templates, and documentation in Spanish, English, and French.

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